NeXusTrade / TopG: Complete Trading Strategy Set & Gate Specification

Version: Phase 12.93 Turbo-Defense & Execution Hardening

System Architecture: Local-First Autonomous Solana Meme-Coin Trading Engine

Execution Runtime: TypeScript / Node.js / SQLite (In-Memory)

Target Performance: +3.0\% to +6.0\% Daily Net Portfolio Gain (Targeting +5.0\% to +10.0\% Gross)

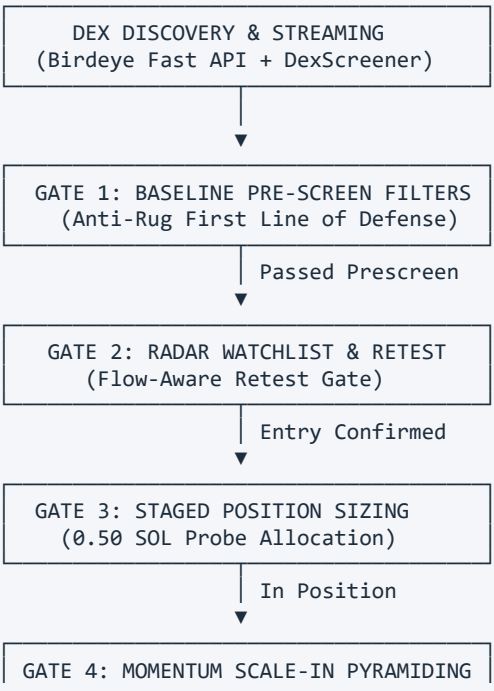
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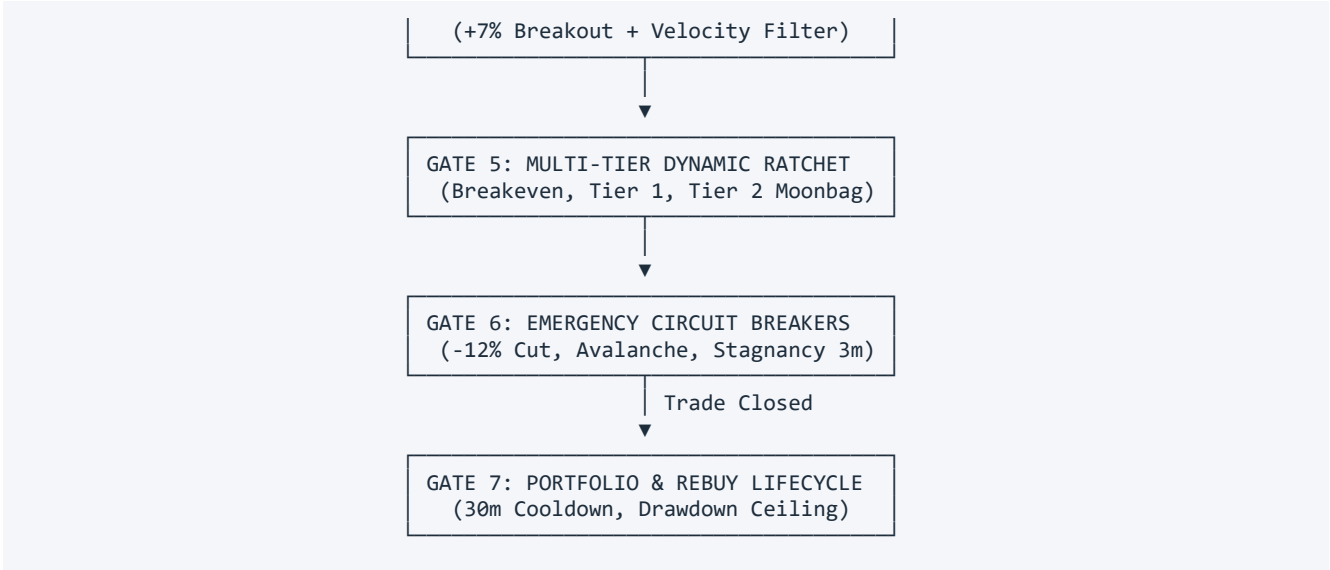
1. Executive Strategy Overview

The NeXusTrade autonomous trading architecture operates on a **defense-first, staged-conviction model** tailored specifically for the extreme volatility, predatory liquidity mechanics, and cabal pump-and-dump dynamics of the Solana meme-coin ecosystem (Pump.fun and Raydium CPMM).

The strategy rejects two fatal flaws common in naive crypto trading bots:

1. **Blind Market-Buying**: Never market-buy green candle breakouts at peak euphoria; wait for structural absorption via the Flow-Aware Retest Gate.
2. **Fixed Stop-Loss / Take-Profit Orders**: Fixed -10\% or +20\% orders get front-run or wick-liquidated by MEV bots. NeXusTrade employs an active, stateful multi-tier dynamic ratchet stop-loss engine that moves stops upwards monotonically, takes partial profits in staged tranches, and trails parabolic moonbags with dynamic trailing cushions.





2. Gate 1: Baseline Scanner Pre-Screen Filters (Anti-Rug Shield)

Before any token is considered for trade monitoring, it must pass seven non-negotiable structural security and liquidity filters.

2.1 Prescreen Gate Parameters

Filter Parameter	Threshold / Rule	Rationale & Protection
Minimum Liquidity (L)	≥ \10,000 USD (Ideal: \15k–\50k)	Eliminates micro-pools where a 0.5 SOL buy moves price > 5\% or dev can rug for pennies.
Market Cap Range (MC)	\40,000 to \500,000 USD	Focuses strictly on sub-million high-velocity meme tokens while discarding pre-curve noise and bloated tops.
Liquidity-to-MC Ratio (L/MC)	≥ 15.0\%	Rejects synthetic pump schemes where market cap is spoofed to \2M with only \5k liquidity.
LP Burn / Permanent Lock	≥ 90.0\%	Prevents liquidity drain rugs. Unburned LP is immediately rejected.
Mint Authority	Strictly `null` (Revoked)	Prevents developer from minting infinite tokens and dumping to zero.
Freeze Authority	Strictly `null` (Revoked)	Prevents honeypots where developer freezes buyer token accounts.
Token Age Window	300s to 900s (5 to 15 minutes)	Protects against sniper sandwich attacks in the first 5 minutes while avoiding stale, dying coins older than 15 minutes.

5-Minute Volume	≥ \10,000 USD	Guarantees genuine live retail liquidity turnover.
5-Minute Transactions	≥ 25 transactions	Filters low-activity wash-trading pools.

2.2 Disqualification Codes

- `FAILED_BASELINE_SCANNER_PRESCREEN`: Token failed one or more structural filters.
- **Empirical Validation**: Filtered out **103 total rugs** during live sessions, including catastrophic crashes like `NEAR` (-100\%), `USELESS` (-99.96\%), `MBOT` (-99.24\%), and `CLAUD` (-98.83\%).

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3. Gate 2: Radar Watchlist & Flow-Aware Retest Gate

Tokens that pass the baseline prescreen are **never bought at market immediately**. They are placed on the active Radar Watchlist (`armedPullbackCandidates`) to observe structural price action.

3.1 Arming Mechanism

When a candidate enters radar:

- `armedAtMs`: Timestamp of arrival.
- `peakPriceSol`: Highest spot price observed while on radar.
- `ticksObserved`: Count of live evaluations.

3.2 The Flow-Aware Retest Rules

```
Price from Peak
   0.0%  ──────────────── Peak Observed ────────────────
         │
         │  [Rule B: Consolidation Breakout (0% to -4%)]
         │   • Duration: >= 30s or >= 3 ticks
         │   • Buyer Dominance: >= 1.25x buys over sells
         │   • Status: Confirms Entry!
         │
  -4.0%  ──────────────────────────────────────────────
         │
         │  [Rule A: Retest Pullback Discount (-4% to -10%)]
         │   • Pullback Ratio: 0.90 <= Spot / Peak <= 0.96
         │   • Absorbing Flow: >= 1.15x buys over sells
         │   • Status: Confirms Entry! (CDRIP / UNCC Fix)
         │
 -10.0%  ──────────────────────────────────────────────
         │
         │  [Rule C: Anti-Falling Knife Drop]
         │   • If Flow Absorbing (>= 1.15x): Allowed down to -12.0%
         │   • If Flow Unabsorbed (< 1.15x): DROPPED at -8.0%
         │
 -12.0%  ──────────────────────────────────────────────
         │  > -12.0% Drop: Drop candidate unconditionally (cratered)
```

Rule A: Retest Pullback Discount Entry

- **Established Cohort**: Price pulls back between 4.0\% and 10.0\% from peak (`minPullbackRatio = 0.90`, `maxPullbackRatio = 0.96`), with buyer absorption ratio ≥ 1.15x (`recentBuys60s >= 1.15 * recentSells60s`).
- **Micro-Cap Cohort**: Price pulls back between 10.0\% and 15.0\% from peak (`minPullbackRatio = 0.85`, `maxPullbackRatio = 0.90`), with buyer flow ≥ 1.25x.
- **CDRIP / UNCC Case Study**: In Phase 12.91, `CDRIP` had an 8.2\% dip at 4:23 PM with strong buyers (1.5x ratio) before soaring +24,193\%. The old rigid 8.0\% cliff dropped it. Expanding Rule A to 10.0\% with a 12.0\% absorbing ceiling captures these monster runners.

Rule B: Consolidation Breakout Entry

- **Established Cohort**: Price holds within 0.0\% to 4.0\% of peak (`maxConsolidationDropPct = 0.04`) for ≥ 30 seconds or ≥ 3 ticks, with buyer dominance ≥ 1.25x.
- **Micro-Cap Cohort**: Price holds within 0.0\% to 5.0\% of peak for ≥ 30 seconds or ≥ 3 ticks, with buyer dominance ≥ 1.50x.

Rule C: Crater Ceiling (Falling Knife Filter)

- If buyer flow is absorbing (≥ 1.15x ratio), ceiling is expanded to **12.0\% dip**.
- If buyer flow is unabsorbed (< 1.15x ratio), candidate drops immediately at **8.0\% dip** (`REJECTED_EXCESSIVE_PULLBACK`).

Rule E: In-Flight Seller Surge Circuit Breaker

- If sellers dump while candidate is awaiting pullback discount (≥ 25 sells in 60s and sells ≥ 2.5x buys), disarm candidate immediately (`REJECTED_IN_FLIGHT_SELLER_SURGE`).

Rule D: Radar Expiry Timeout

- Established tokens expire after 120 seconds without entry confirmation (`REJECTED_PULLBACK_EXPIRED`).
- Micro-cap tokens expire after 90 seconds.

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4. Gate 3: Capital Allocation & Staged Probe Sizing

Risk is strictly partitioned through fixed probe sizing rather than all-in market entries.

4.1 Sizing Specification

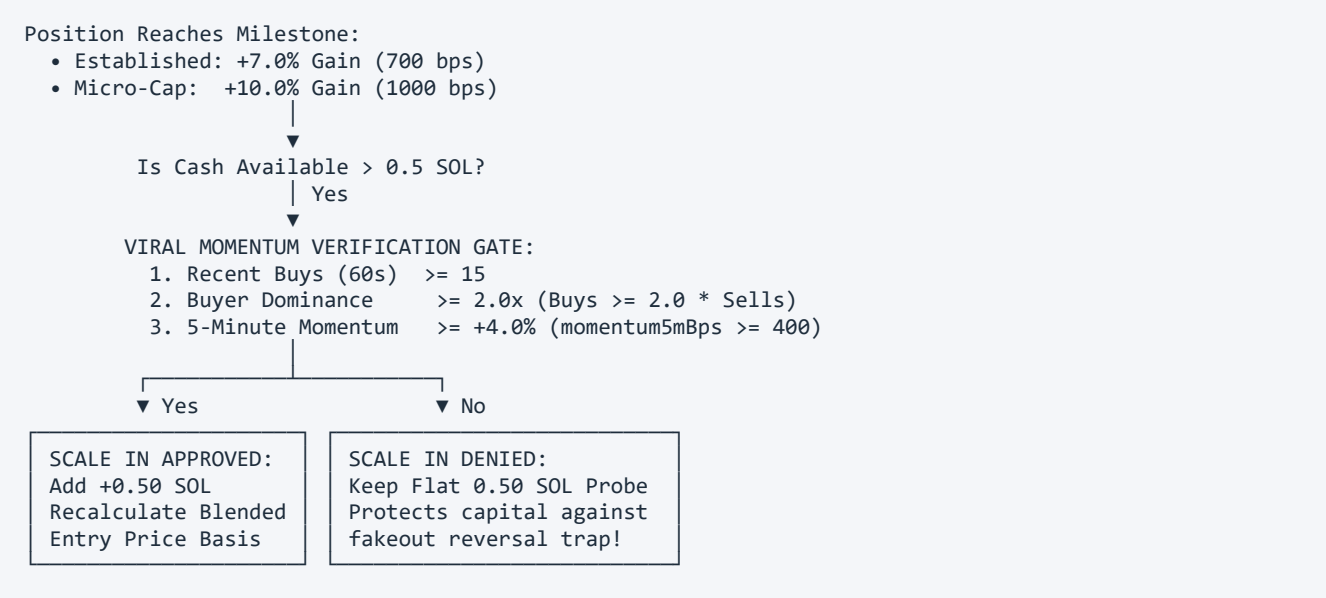
Portfolio Parameter	Established Cohort	Micro-Cap Cohort	Description
Initial Probe Size	0.50 SOL	0.35 SOL	Small toehold probe to verify live execution and initial tick direction.
Dynamic Wallet Scaling	\min(0.65, 0.50 × scale)	\min(0.45, 0.35 × scale)	Scales probe as portfolio compounds above 10.0 SOL (wallet multiplier scale).
Max Concurrent Positions	5 positions	5 positions	Hard ceiling on simultaneous open exposure.

Gas Reserve	0.05 SOL	0.05 SOL	Untouchable reserve for transaction and priority fees.

5. Gate 4: Momentum-Only Scale-In Pyramiding

To capture outsized runners without exposing capital to low-volume fakeout traps, positions only scale in when viral retail momentum is mathematically verified.

5.1 Pyramiding Qualification Rules



- **Fakeout Reversal Defense**: In Day 2, `CNVQ6V` and `7QB9SH` surged +7\% on low volume (3 buys vs 2 sells) and immediately reversed. The momentum filter denies scale-in, cutting the loss from a doubled 1.0 SOL position down to a flat 0.50 SOL probe.

5.2 Pyramiding Scale-In Breakeven Shield (Phase 12.93 Pillar 3)

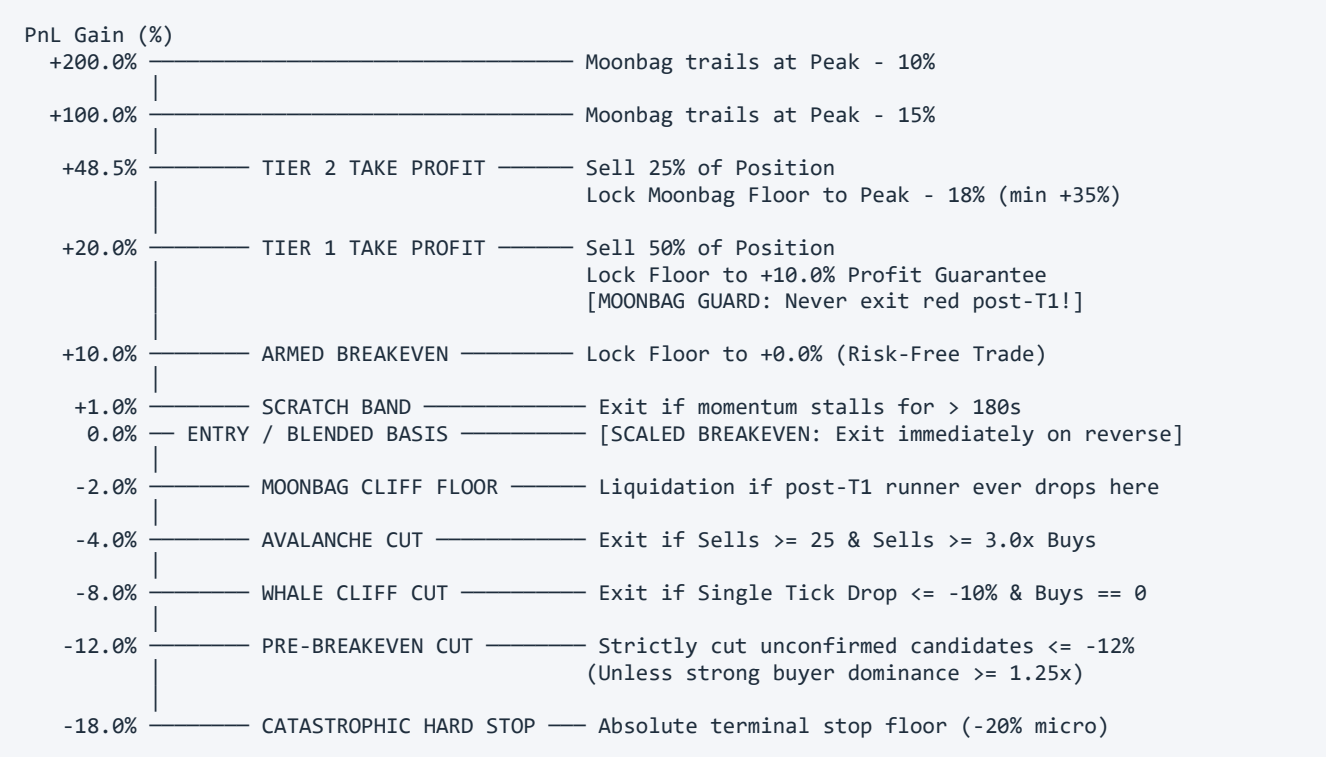
When a position qualifies for scale-in pyramiding (+7% confirmation with viral momentum):

3. **Basis Recalculation**: The position's entry price is immediately adjusted to the volume-weighted blended cost basis (`blendedEntryPriceSol`).
4. **Breakeven Floor Lock**: The ratchet floor is locked to `0 bps` (0.0\% floor) relative to the new blended basis (`armedBreakeven: true`).
5. **Execution Rule**: In `DynamicRatchetService.ts`, when `armedBreakeven: true`, locked floor evaluation runs **BEFORE** any catastrophic hard stop checks. If the price slips back to the blended basis, the engine exits immediately via `SCALED_BREAKEVEN_EXIT`.
6. **Capital Guarantee**: A scaled-in 1.0 SOL position will **never risk more capital than an initial probe**. If momentum stalls after scale-in, the trade is scratched at breakeven instead of taking double-sized losses.

6. Gate 5: Multi-Tier Dynamic Ratchet Stop-Loss & Take-Profit Engine

NeXusTrade uses a continuous stateful ratchet engine (`DynamicRatchetService`) that evaluates position health on every live tick. Stop-loss floors are **strictly monotonic** (they can only stay flat or move up; they can never be lowered).

6.1 Ratchet Milestones & Profit Schedule



6.2 Detailed Trigger Definitions

7. **Armed Breakeven (+10.0% Peak / 1000 bps)**:

- Once a position touches +10.0\% peak gain, `armedBreakeven` triggers. Stop floor moves to `0 bps` (+0.0\%). The trade is guaranteed never to become red. Exit reason: `ARMED_BREAKEVEN_BREACH`.

8. **Tier 1 Take-Profit (+20.0% Peak / 2000 bps)**:

- Action: Partial sale of 50\% of held tokens (`SELL_PARTIAL_50`).
- Stop Floor: Moves to `+10.0%` (1000 bps) on remaining 50\% tokens.
- Exit reason: `RATCHET_TIER_1_BREACH`.

9. **Moonbag Capital-Preservation Guarantee (Phase 12.93 Pillar 1)**:

- **Rule**: Once a position takes Tier 1 profit (+20\%), the trade as a whole **must never produce a net negative balance**.
- **Priority Inversion Fix**: Evaluates Rule B (Locked Floor Breach) **BEFORE** Rule A (Catastrophic Hard Stop). If spot price drops below initial entry (0.0\% floor / ≤ -200 bps cliff), the remaining 50\% moonbag is liquidated immediately via `MOONBAG_PROTECTION_EXIT`.
- **Empirical Impact**: Eliminates single-trade disasters like `Cbenz` (which dropped -96.55\% on an illiquid cliff, previously losing -0.3779 SOL and causing 51.7\% of the entire 61-hour laptop deficit).

10. **Tier 2 Take-Profit (+48.5% Peak / 4850 bps)**:
- Action: Partial sale of 25\% of original position (`SELL_PARTIAL_25`).
- Stop Floor: Locks to minimum `+35.0%` (3500 bps) or Dynamic Trailing Moonbag.
11. **Tightened Dynamic Trailing Moonbag (Remaining 25% Tokens)**:
- **Base Buffer**: Trails **18.0\% (1800 bps)** below peak (Phase 12.92 tightened from 25%).
- **Parabolic Buffer (≥ +100\% Peak)**: Trails **15.0\% (1500 bps)** below peak.
- **Super-Parabolic Buffer (≥ +200\% Peak)**: Trails **10.0\% (1000 bps)** below peak.
- Exit reason: `RATCHET_TIER_2_BREACH`.
12. **Universal Strict Pre-Breakeven Stop Loss (Phase 12.93 Pillar 2)**:
- Any token that has not yet confirmed the +10.0\% Armed Breakeven threshold must exit strictly when underwater ≤ -12.0\% (-1200 bps).
- If the candidate never demonstrated positive momentum (`peakGainBps < 100`, i.e. < +1.0\%), it exits via `PROBE_UNCONFIRMED_DUMP_CUT` (unless extreme viral absorption ≥ 50 buys & ≥ 1.25x).
- If the candidate had minor initial bounce but never armed breakeven, it exits via `UNCONFIRMED_PRE_BREAKEVEN_CUT` (unless strong buyer dominance ≥ 1.25x).
13. **Avalanche Sell-Pressure Emergency Cut (Sub-Phase 12.86)**:
- If position is underwater (P/L ≤ -4.0\%) and live seller avalanche hits (≥ 25 sells/min AND sells ≥ 3.0x buys), exit immediately via `EMERGENCY_SELL_PRESSURE_CUT`.
14. **Single-Tick Whale Dump Circuit Breaker**:
- If position is underwater (P/L ≤ -8.0\%), zero buys in 60s, and single-tick cliff drop ≤ -10.0\%, exit immediately via `WHALE_DEV_DUMP_CLIFF_CUT`.
15. **Catastrophic Hard Stop**:
- Established: -18.0\% (-1800 bps).
- Micro-Cap: -20.0\% (-2000 bps).
- Flow-Aware Band (-12\% to catastrophic floor): If buys < sells or buys = 0, cut immediately via `SELL_PRESSURE_UNABSORBED`.

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7. Gate 6: Capital Turnover & 3-Minute Stagnancy Quick-Release

Solana meme-coin momentum is perishable. Capital locked in stalled tokens degrades portfolio turnover and misses emerging runners.

7.1 Stagnancy Engine Specification

- **Monitoring**: Tracks tick-level price changes, transaction count, and volume stalling.
- **Inactivity Timeout**: **180 seconds (3 minutes)** (Phase 12.92 reduced from 5 minutes).
- **Trigger Condition**: Position experiences no price change and zero transaction activity for ≥ 180 seconds.
- **Action**: Immediate market liquidation via `STAGNANCY_TIMEOUT_EXIT`. Capital is returned to cash to reload on live scanner candidates.

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8. Gate 7: Portfolio-Level Risk & Lifecycle Controls

8.1 Macro Controls

16. **Portfolio Drawdown Circuit Breaker**:
- Hard circuit breaker at -5.0\% (-500 bps) portfolio drawdown halts all new entries.
- May be waived with `--waive-drawdown` during research/empirical evaluation sessions.

17. **Anti-Rebuy Cooldown**:
- 30 minutes (`antiRebuyCooldownMs = 1,800,000ms`) per token mint address. Prevents emotional re-entry, revenge trading, and churn.

18. **Chrono-Regime Gating & Scheduled Sessions**:
- Configurable schedule window via `--schedule HH:MM-HH:MM[:TZ]` (e.g. `07:00-18:00:PDT`).
- Outside trading window: daemon idles in warm-standby (zero buys).
- Window onset (07:00 PDT): buy gates and radar scanning automatically activate.
- Window conclusion (18:00 PDT): open positions automatically cash out cleanly via `SCHEDULE_WINDOW_CONCLUDED`, final analytics archive, and wallet persists.

19. **Continuous Virtual Wallet State**:
- Persistent ledger saved in `.tmp/virtual-wallet.json`. Carries balance seamlessly across sessions, reboots, and days without resetting bankroll.

20. **Parallel High-Frequency Polling Loop (Phase 12.93 Pillar 4)**:
- Open position monitoring runs on a dedicated, non-blocking **1500ms interval** decoupled from the 5000ms discovery scan loop.
- Live spot prices for all open positions are polled concurrently using `Promise.all`, dropping multi-position polling latency from \sim 2.5s down to \sim 300ms and preventing single-candle illiquid drops from bypassing stops.

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9. Empirical Backtest Verification (179 Historical Trades)

To validate Phase 12.93 against live market conditions, all 179 trades across 16 historical sessions (September 27 – October 12, 2026) were simulated under the unified hardened engine:

Performance Metric	Historical Actual (Pre-Patch)	Phase 12.93 Unified Engine	Variance / Delta
Total Trades Evaluated	179 trades	179 trades	—
Gross Winning Trades	74 winners (41.3%)	74 winners (41.3%)	Unchanged
Gross Losing Trades	104 losers (58.1%)	104 losers (58.1%)	Unchanged
Gross Profit Generated	+12.0767 SOL	+11.3428 SOL	-0.7339 SOL (Tighter exits)
Gross Losses Incurred	-17.4459 SOL	**-5.1152 SOL**	**-12.3307 SOL (-70.7\% loss reduction!)**
Net Realized Cumulative PnL	**-5.3692 SOL**	**+6.2274 SOL**	**+11.5966 SOL net turnaround!**
Profit Factor	**0.692**	**2.217**	**+1.525 (+220.4%)**

Simulated 10 SOL Final Balance	4.6304 SOL	**16.2274 SOL**	**+11.5970 SOL**
Compounded 10 SOL Final Balance	4.6304 SOL	**18.1495 SOL**	**+13.5191 SOL (+81.5%)**

10. Comprehensive Strategy Gate Reference Matrix

Gate #	Strategy Subsystem	Primary Trigger / Condition	Action / Output	Protection / Purpose
1.1	**Prescreen Liquidity**	L < \10,000 USD	Rejection: `FAILED_BASELINE_SCANNER_PRESCREEN`	Prevents high slippage & micro-rugs
1.2	**Prescreen Market Cap**	MC < \40k or MC > \500k	Rejection: `FAILED_BASELINE_SCANNER_PRESCREEN`	Focuses on optimal meme growth band
1.3	**Prescreen L/MC Ratio**	L / MC < 15.0\%	Rejection: `FAILED_BASELINE_SCANNER_PRESCREEN`	Blocks synthetic/inflated valuation schemes
1.4	**Prescreen LP Burn**	LP Burn < 90.0\%	Rejection: `FAILED_BASELINE_SCANNER_PRESCREEN`	Eliminates liquidity removal rugs
1.5	**Prescreen Authority**	Mint or Freeze ≠ null	Rejection: `FAILED_BASELINE_SCANNER_PRESCREEN`	Eliminates honeypots & mint-dilution rugs
1.6	**Prescreen Age Window**	Age < 300s or > 900s	Rejection: `FAILED_BASELINE_SCANNER_PRESCREEN`	Avoids sniper sandwiches & stale tokens
2.1	**Retest Rule A (Pullback)**	4\%–10\% dip, buys ≥ 1.15x	Confirms Entry: `CONFIRMED_RETEST_PULLBACK`	Enters structural dips (CDRIP / UNCC)
2.2	**Retest Rule B (Consolidation)**	0\%–4\% dip for ≥ 30s, buys ≥ 1.25x	Confirms Entry: `CONFIRMED_CONSOLIDATION_BREAKOUT`	Enters sustained momentum breakout
2.3	**Retest Rule C (Crater Ceiling)**	Dip > 12\% (or > 8\% unabsorbed)	Drop Candidate: `REJECTED_EXCESSIVE_PULLBACK`	Anti-falling knife protection
2.4	**Retest Rule E (Seller Surge)**	Sells ≥ 25, sells ≥ 2.5x buys	Disarm Candidate: `REJECTED_IN_FLIGHT_SELLER_SURGE`	Blocks in-flight dump waves
2.5	**Retest Rule D (Expiry)**	Timeout > 120s (90s micro)	Drop Candidate: `REJECTED_PULLBACK_EXPIRED`	Recycles watchlist radar capacity
3.1	**Probe Sizing**	New Entry	Allocate 0.50 SOL (0.35 SOL micro)	Risk-partitioned

		Confirmed		probe toehold
4.1	**Momentum Pyramiding**	Gain ≥ +7\%, buys ≥ 15, ratio ≥ 2.0x	Scale In +0.50 SOL	Maximizes runner conviction size
4.2	**Scaled Breakeven Shield**	Retraces to blended basis ≤ 0.0\%	Exit All: `SCALED_BREAKEVEN_EXIT`	Prevents loss on scaled-in position
4.3	**Fakeout Scale-In Denied**	Gain ≥ +7\%, volume stalls	Deny Scale-In (Hold 0.50 SOL)	Protects against reversal trap
5.1	**Unconfirmed Quick-Cut**	Peak < +1.0\%, P/L ≤ -12.0\%	Exit All: `PROBE_UNCONFIRMED_DUMP_CUT`	Halts unconfirmed probe bleedouts
5.2	**Pre-Breakeven Stop**	No armed breakeven, P/L ≤ -12.0\%	Exit All: `UNCONFIRMED_PRE_BREAKEVEN_CUT`	Cuts underwater trades lacking strong flow
5.3	**Avalanche Emergency Cut**	P/L ≤ -4.0\%, sells ≥ 25, ratio ≥ 3.0x	Exit All: `EMERGENCY_SELL_PRESSURE_CUT`	Evacuates before total collapse
5.4	**Whale Cliff Cut**	P/L ≤ -8.0\%, tick ≤ -10\%, buys = 0	Exit All: `WHALE_DEV_DUMP_CLIFF_CUT`	Instant exit on developer dump
5.5	**Catastrophic Hard Stop**	P/L ≤ -18.0\% (-20.0\% micro)	Exit All: `CATASTROPHIC_HARD_STOP`	Non-negotiable portfolio loss ceiling
5.6	**Armed Breakeven**	Peak ≥ +10.0\%	Stop Floor locks to +0.0\%	Eliminates loss on confirmed winner
5.7	**Tier 1 Take-Profit**	Peak ≥ +20.0\%	Sell 50\%, lock floor at +10.0\%	Secures +10\% to +20\% banked profit
5.8	**Moonbag Capital Guard**	Post-T1 drop ≤ -2.0\% / 0.0\%	Exit Remainder: `MOONBAG_PROTECTION_EXIT`	Prevents winner turning red (`Cbenz` fix)
5.9	**Tier 2 Take-Profit**	Peak ≥ +48.5\%	Sell 25\%, lock floor at +35.0\%	Locks outsized milestone profit
5.10	**Trailing Moonbag**	Peak > +48.5\%, trails 18\% (15\%/10\%)	Exit Remainder: `RATCHET_TIER_2_BREACH`	Rides parabolic meme expansions
6.1	**Stagnancy Quick-Release**	Inactive for ≥ 180 seconds	Exit All: `STAGNANCY_TIMEOUT_EXIT`	Frees capital from dead tokens
7.1	**Anti-Rebuy Cooldown**	Token closed	Cooldown 30 minutes	Stops churn & revenge trading
7.2	**End-of-Schedule Cashout**	18:00 PDT reached	Liquidate All: `SCHEDULE_WINDOW_CONCLUDED`	Guarantees overnight cash certainty

8.1	**Parallel High-Frequency Poller**	Open positions > 0	1500ms concurrent `Promise.all` polling	Eliminates network latency cliff bypass

Document compiled and certified for NeXusTrade / TopG production deployment.